AI and neuroimaging meta-analysis

CHIA-SHU LIN

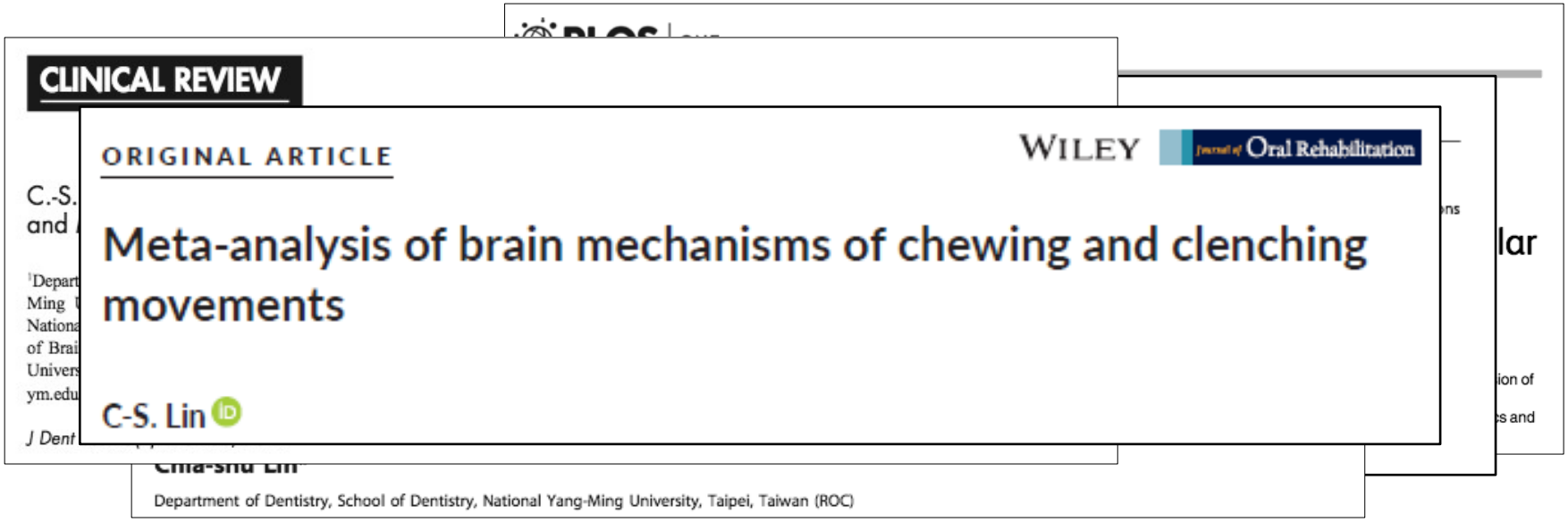
Department of Dentistry
National Yang Ming Chiao Tung University, TAIWAN (ROC)

About me

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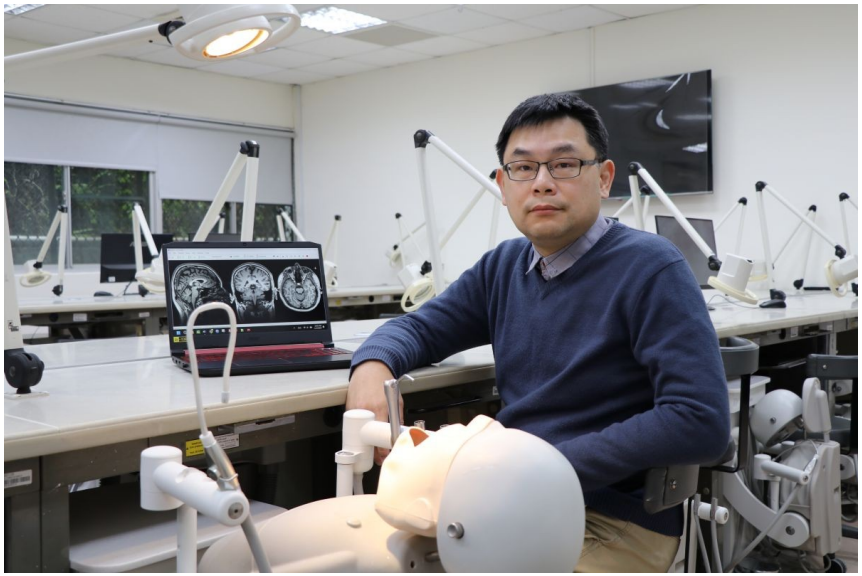
AI meta-analysis ~

2011. University of Oxford, Oxford Centre for Functional MRI of the Brain (FMRIB), DPhil Anaesthesia
2006. Imperial College London, MSc
2000. National Yang-Ming University, DDS



You *should* write reviews

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Ageing Research Reviews 93 (2024) 102128

Contents lists available at ScienceDirect

Ageing Research Reviews

journal homepage: www.elsevier.com/locate/arr

Review article

An umbrella review on the association between factors of oral health and cognitive dysfunction

Chia-Shu Lin [a,b,c,d,*], Ta-Chung Chen [e], Merel Charlotte Verhoeff [f], Frank Lobbezoo [f], Mats Trulsson [g,h], Jong-Ling Fuh [b,i,j]

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[b] Institute of Brain Science, National Yang Ming Chiao Tung University, Taiwan
[c] Brain Research Center, National Yang Ming Chiao Tung University, Taiwan
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[f] Department of Orofacial Pain and Dysfunction, Academic Centre for Dentistry Amsterdam (ACTA), University of Amsterdam and Vrije Universiteit Amsterdam, Amsterdam, the Netherlands
[g] Division of Oral Diagnostics and Rehabilitation, Department of Dental Medicine, Karolinska Institutet, Huddinge, Sweden
[h] Academic Center for Geriatric Dentistry, Karolinska Institutet, Stockholm, Sweden
[i] Department of Neurology, Neurological Institute, Taipei Veterans General Hospital, Taipei, Taiwan
[j] School of Medicine, College of Medicine, National Yang Ming Chiao Tung University, Taipei, Taiwan

Go systematic

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50%
paper

12 (
5)

r=0.9 (
)

Lancet NEJM

12
24

(p r
)

Definition

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- Systematic review[1]

‘A systematic review attempts to collate all empirical evidence that fits pre-specified eligibility criteria **to answer a specific research question**.

It uses **explicit, systematic methods** that are selected with a view to minimizing bias, thus providing reliable findings from which conclusions can be drawn and decisions made. The key characteristics of a systematic review are: [1]

- Meta-analysis[1]

‘Meta-analysis is the **use of statistical techniques** to integrate and summarize the results of included studies.’[1]

(p r)

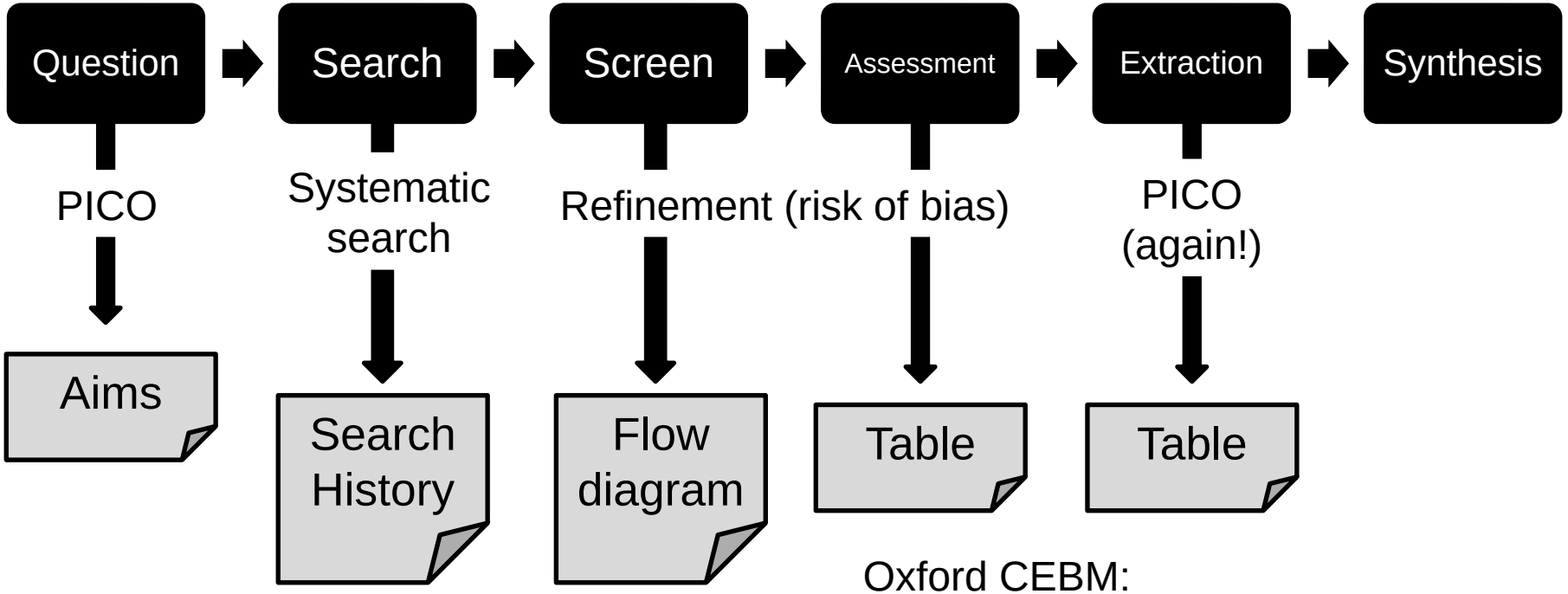
[1]Liberati et al. (2009) *PLOS MED*

PRISMA: the concept

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- PRISMA: **P**referred **R**eporting **I**tems for **S**ystematic reviews and **M**eta-**A**nalyses

http://www.prisma-statement.org/

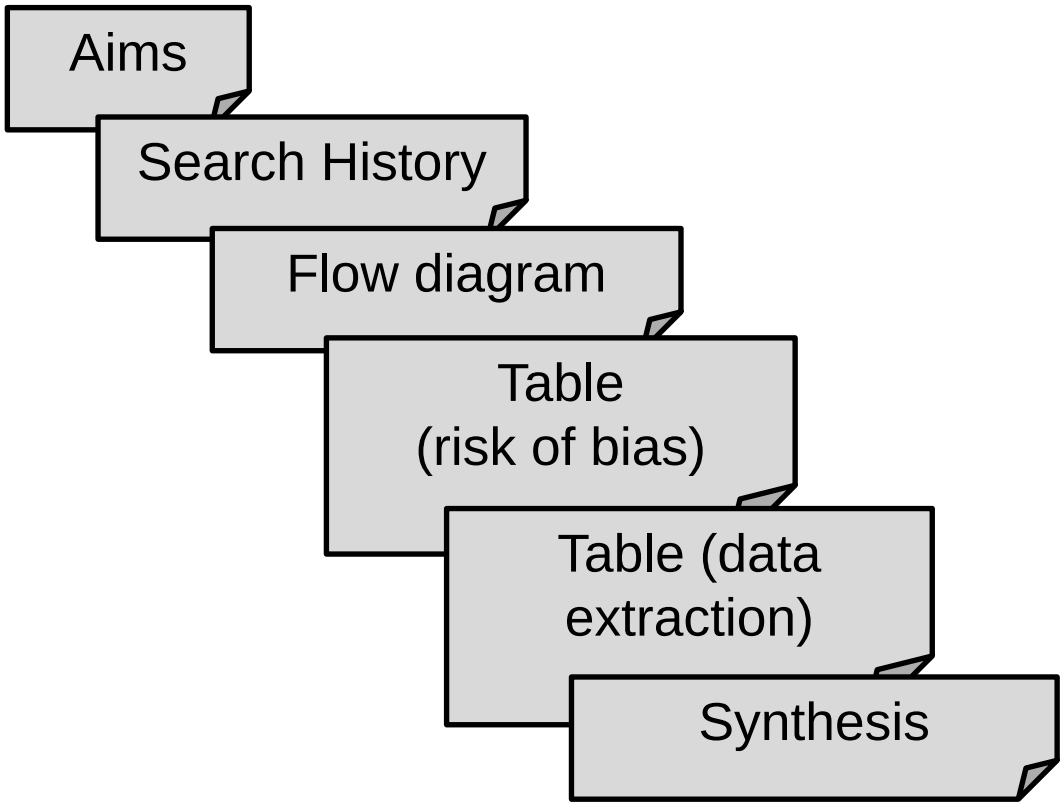


Oxford CEBM:
https://www.cebm.ox.ac.uk/resources/ebm-tools/finding-the-evidence-tutorial

PRISMA: an example

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■ Cunha-Cruz et al. (2010) Dentin Hypersensitivity and Oxalates: a Systematic Review, *J Dent Res*



Systematic search

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- Search history

History

Download history Clear history

Search	Add to builder	Query	Items found	Time
#7	Add	Search **#6 NOT (Case Reports[ptyp] OR Review[ptyp])** Filters: **Journal Article; Publication date from 1994/01/01 to 2013/08/31**	81	23:51:02
#6	Add	Search **(#1 OR #2) AND #3** Filters: **Journal Article; Publication date from 1994/01/01 to 2013/08/31**	182	23:51:02
#5	Add	Search **(#1 OR #2) AND #3** Filters: **Journal Article**	192	23:49:20
#4	Add	Search **(#1 OR #2) AND #3**	195	23:49:12
#3	Add	Search **pain (central nervous system OR brain) MRI**	5410	23:49:12
#2	Add	Search **temporomandibular disorder**	14781	23:49:12
#1	Add	Search **trigeminal (neuralgia OR neuropathy OR neuropathic)**	7355	23:49:12

Source: Chia-Shu Lin

((trigeminal AND (neuralgia OR neuropathic)) OR (temporomandibular disorder)) AND (pain AND (central nervous system) OR brain) AND MRI)

Systematic search

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- Eligibility criteria

Study characteristics	Report characteristics
What are the objectives of the studies	In what way are the studies presented?
P/I/C/O	language of publication, publication status (e.g., inclusion of unpublished material and abstracts), and year of publication

–

Systematic search

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- ‘tags' and 'fields’

Fields	Tag	Examples
Author	[au]	Death[au]
Affiliation	[ad]	"National Yang-Ming University"[ad]
Journal Title	[ta]	JADA[ta]
Publication Type	[pt]	“case reports”[pt] review[pt]

- https://www.ncbi.nlm.nih.gov/books/NBK3827/table/pubmedhelp.Tn

- Wildcards: *

masticat* (masticatory, mastication, masticator…)
cognit* (cognition, cognitive)
dement* (demented, dementia)

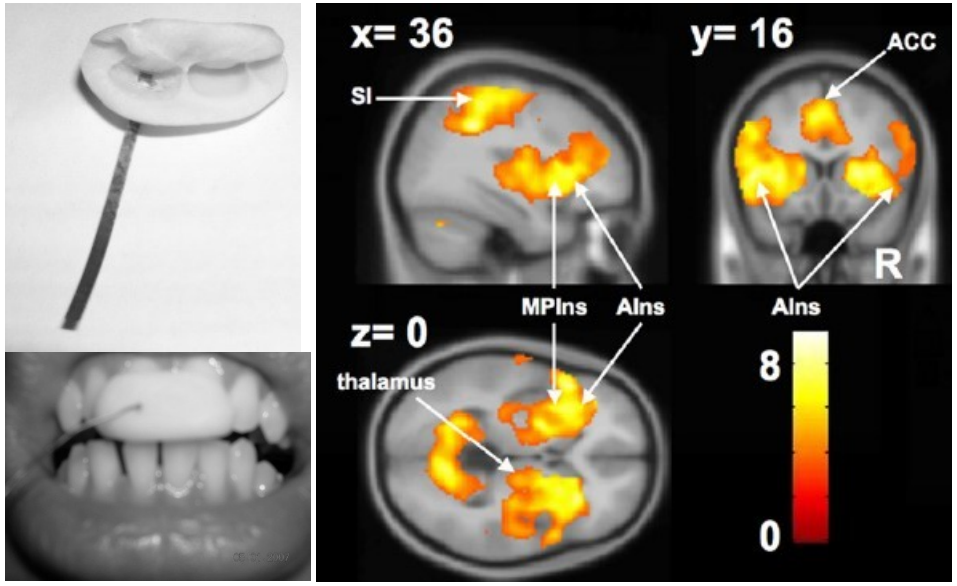
AI

Neuroimaging meta-analysis

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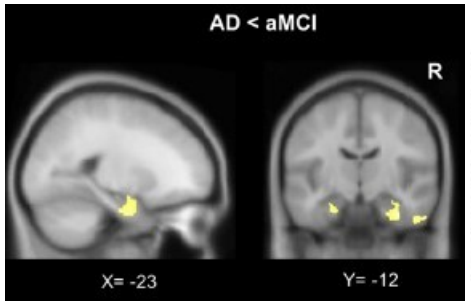
- Neuroimaging: structural vs. functional brain features

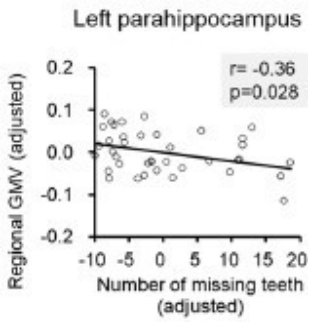
Task-based functional MRI[1]



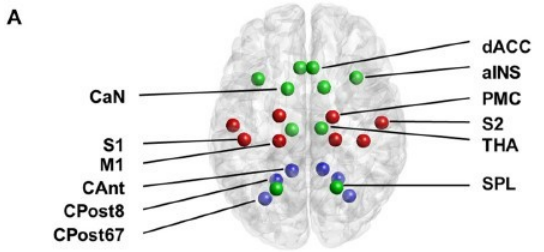
*Blood-oxygen-level-dependent imaging

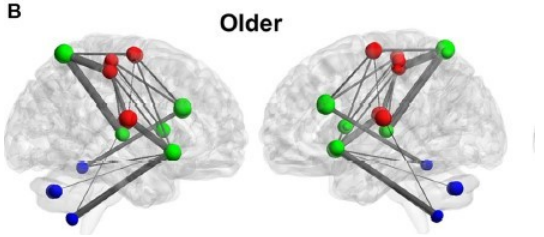
Structural MRI[2]





Resting-state functional MRI[3]



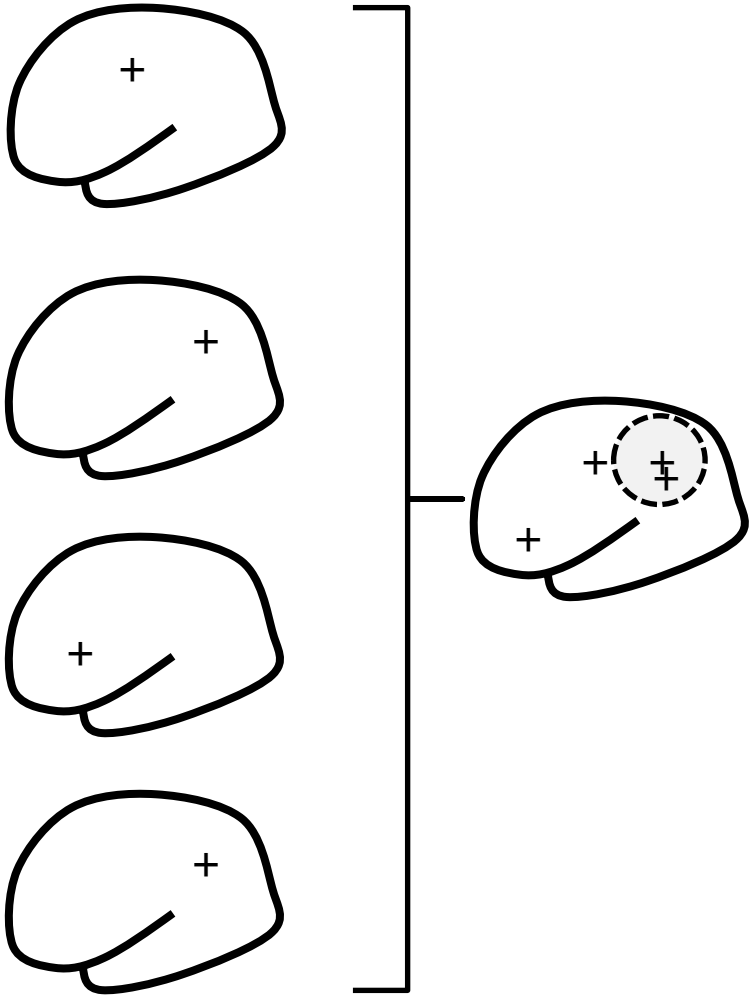


[1]Lin et al. (2013) *Brain Research*
[2]Lin et al. (2020) *Brain Imaging and Behavior*
[3]Lin et al. (2017) *Frontiers in Aging Neuroscience*

Neuroimaging meta-analysis

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- (location)



whole-brain analysis(bias)

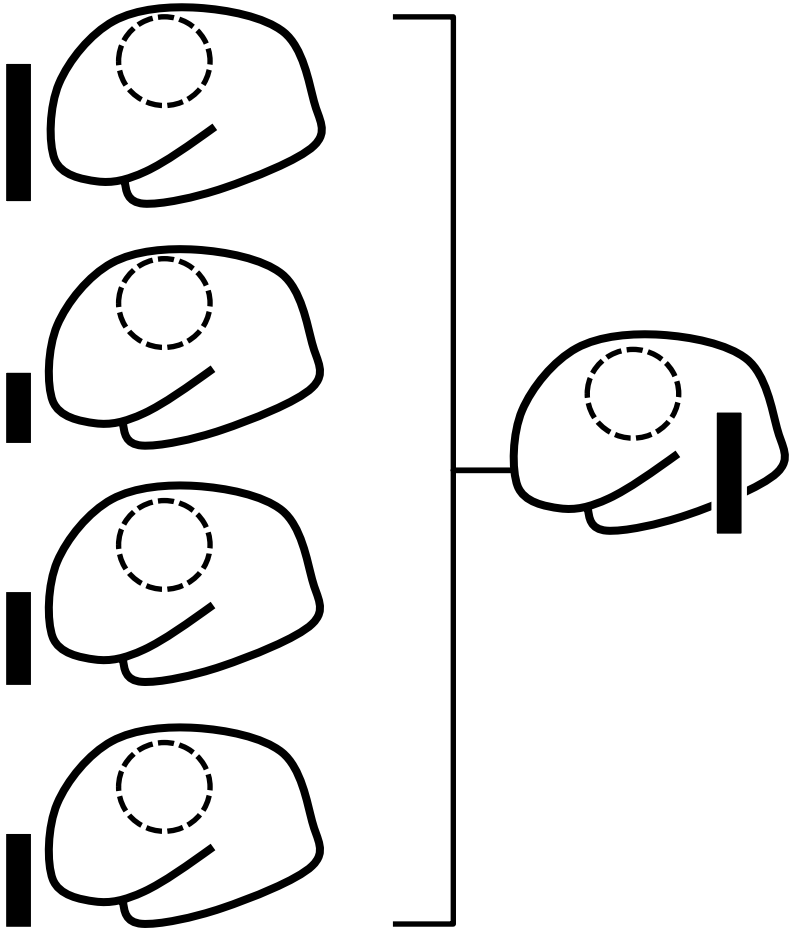
voxel study
cluster?

Pooled effect ( )
voxel activation?

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- / (intensity)



ROI analysis ( )

voxel study
activation ?

Pooled effect ?

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- Data synthesis: Coordinate-based analysis[1]

- Label-based reviews
  -
  - parecellation scheme (AAL)
  - (peak coordinates)
  - study fixed effect analysis

- Activation Likelihood Estimation (ALE)
  - study peak coordinate
  - Data synthesis
  -

[1]Radua & Mataix-Cols (2012) *Biology of Mood & Anxiety Disorders*

Activation Likelihood Estimation

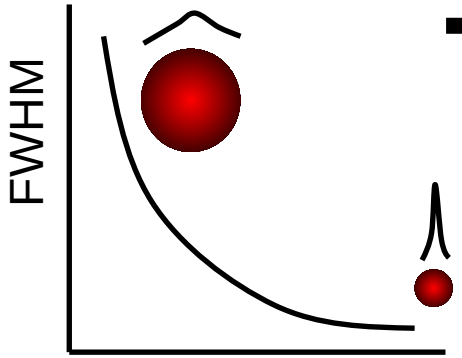
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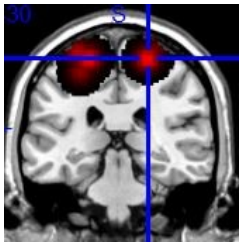
loci coordinates
TAL MNI)

```
//Reference=MNI
//Study 1
//Subjects=20
-22     -30     65
22      -30     65
-27     -22     71
27      -22     71
-30     -35     70
-28     -37     55

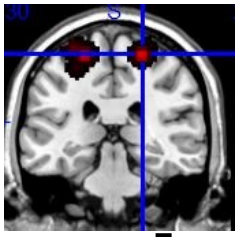
//Study 2
//Subjects=200
27      -22     71
30      -35     70
22      -30     65
```

loci



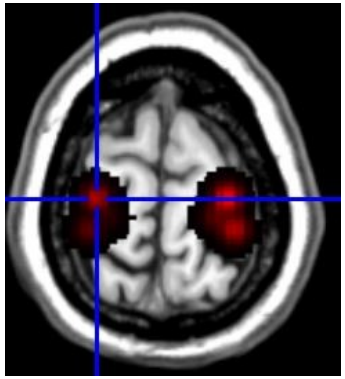


study1



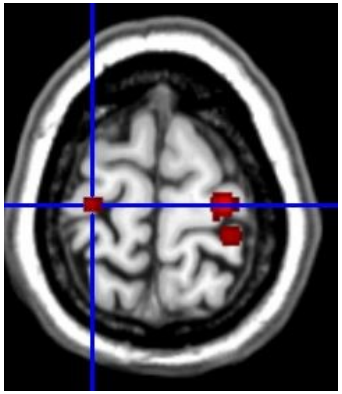
study2

ALE map

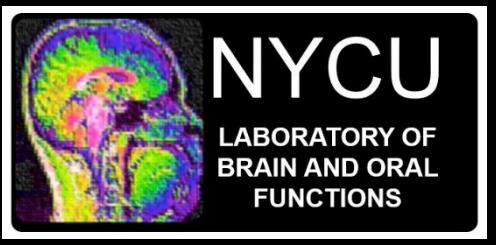


Statistical thresholding

Final outcome



AI-enhanced meta-analysis



- Neurosynth

- ‘NeuroSynth combines text mining, meta-analysis, and machine learning techniques to generate probabilistic mappings between cognitive and neural states that can be used for a broad range of neuroimaging applications.’[1]

- ASreview (https://asreview.nl)

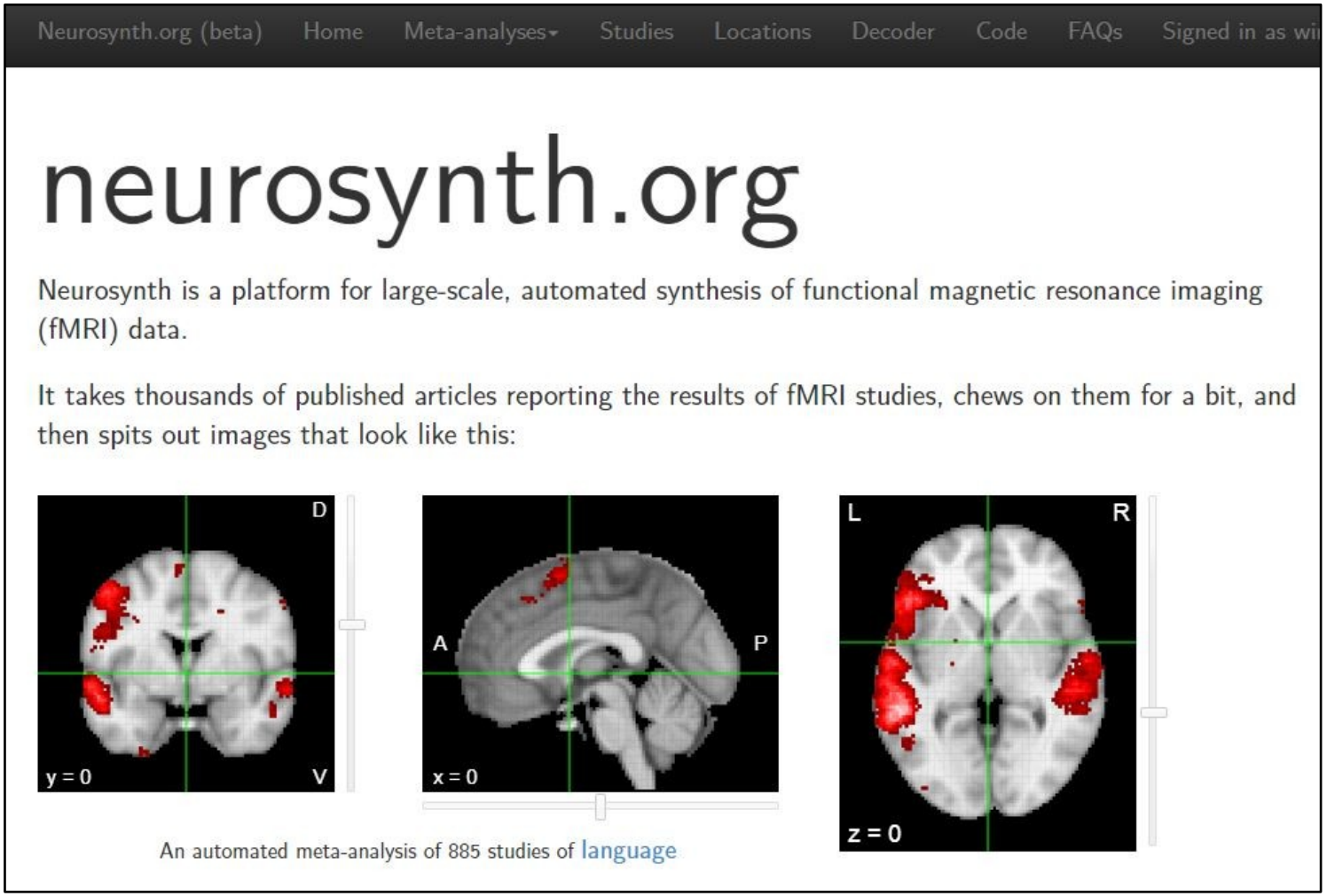
- ‘The future of systematic reviewing will be an interaction with machine learning algorithms to deal with the enormous increase of available text. … We demonstrate by means of simulation studies that active learning can yield far more efficient reviewing than manual reviewing while providing high quality’[2]

[1]Yarkoni et al. 2011 Nature Methods
[2]de Schoot et al. (2021) Nature Machine Intelligence

Automatic synthesis

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- Large-scale, automated synthesis (Neurosynth)



Neurosynth

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Forward inference: P(Activation|Term)

The likelihood to find the locus activated given that this term is investigated

Reverse inference: P(Term|Activation)

The likelihood that the term is associated given that the locus is found activated.

$$P(\text{Term/Activation}) = \frac{P(\text{Activation/Term}) \times P(\text{Term})}{P(\text{Activation})}$$

$$P(\text{Term/Activation}) = P(\text{Activation/Term})$$

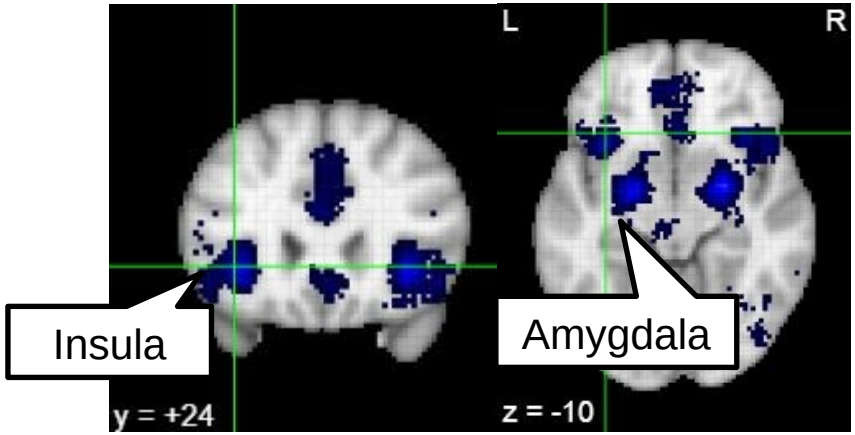
Fallacy of reverse inference

Fallacy P(Activation)
^^

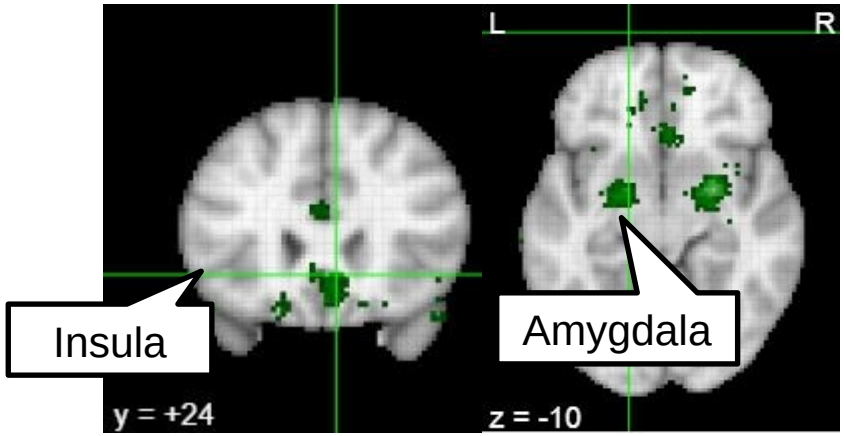
Neurosynth: an example

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- Term= ‘fear’



Uniformity map: consistent activation



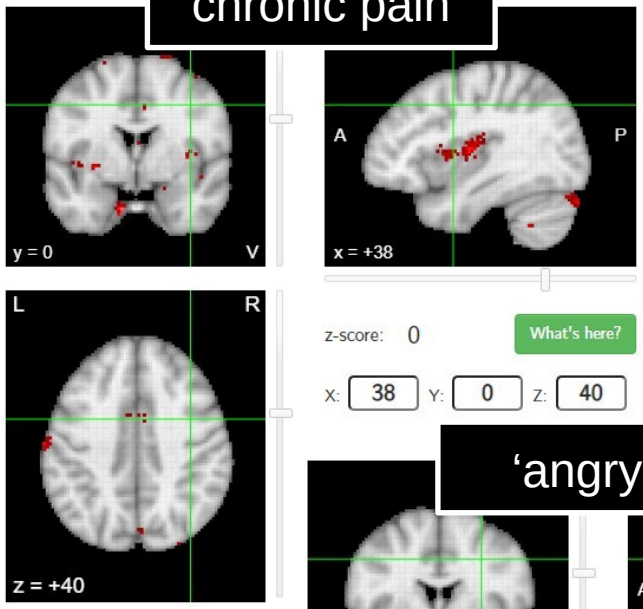
Association map: preferential activation

Insula (fear insular activation)

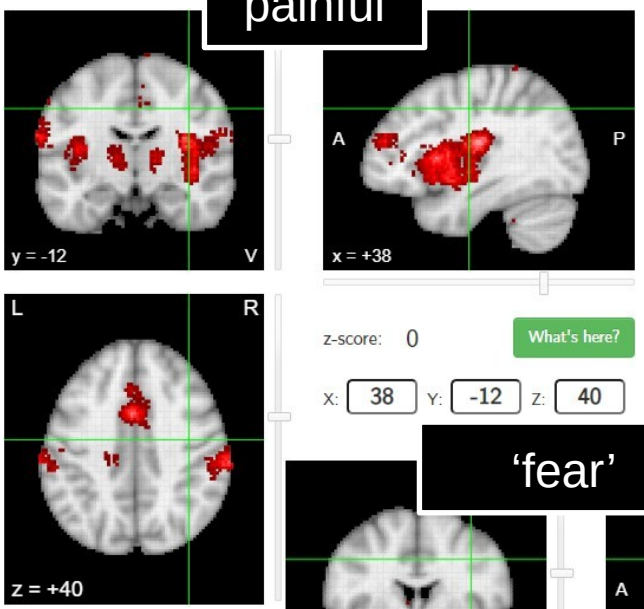
Beware of a *simple* brain theory

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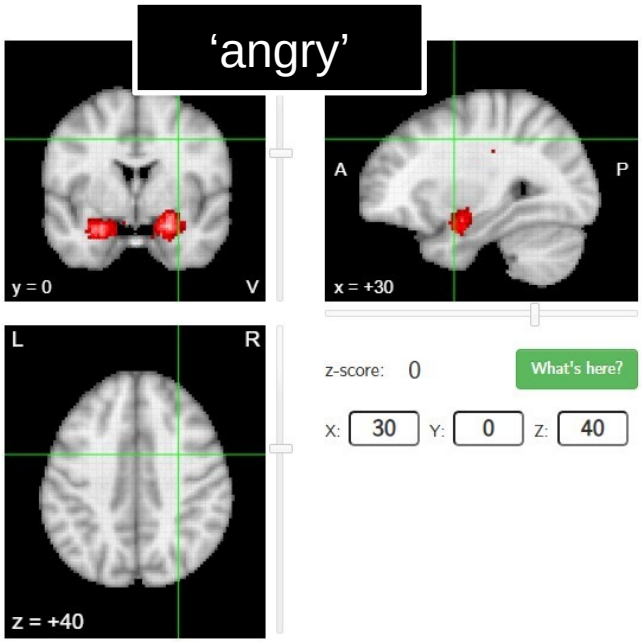
‘chronic pain’



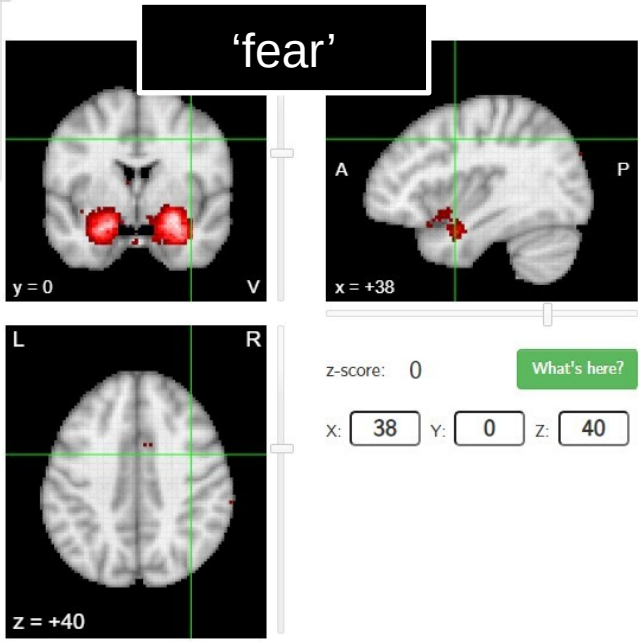
‘painful’



‘angry’

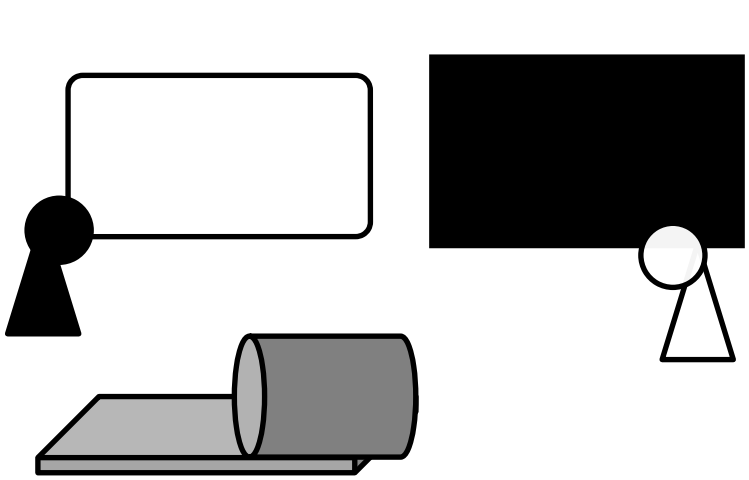


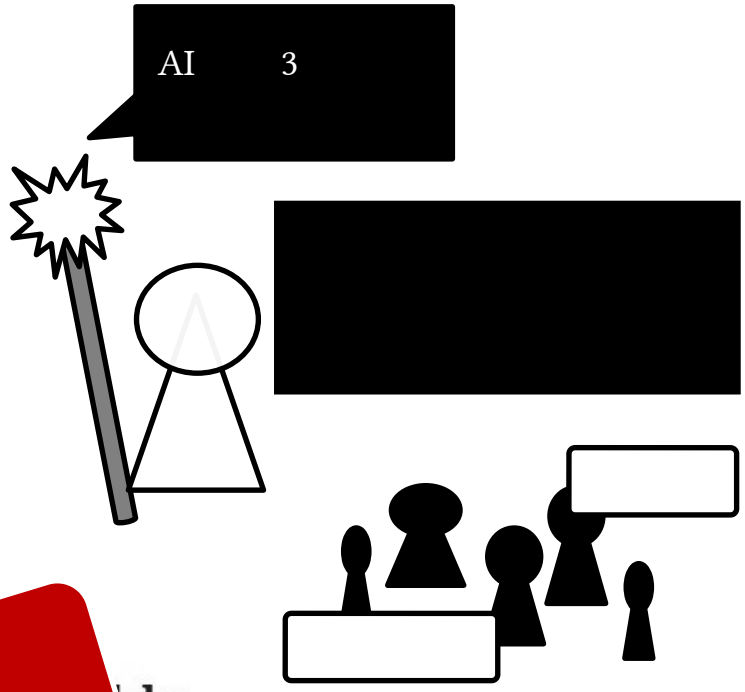
‘fear’



Clinical and ethical issues

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Review > Neuron. 2020 Mar 4;105(5):771-775. doi: 10.1016/j.neuron.2020.01

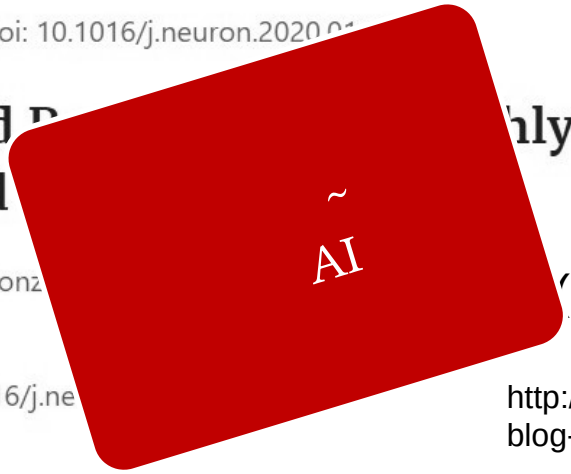
Ethical Issues Posed by Field ... hly Portable and Cloud-Enabled ...

Francis X Shen [1], Susan M Wolf [2], Ramon Gilberto Gonz...

Affiliations + expand

PMID: 32135089 PMCID: PMC8803403 DOI: 10.1016/j.ne...

Free PMC article



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(2017)

http://scimonth.blogspot.com/2017/03/blog-post_24.html